The zero locus of $\Psi^C_{(1^r),\theta}$ in M: reciprocal pairs are a *restriction* locus, not a vanishing locus

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Provenance (PROTOCOL §2.3).
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Repository: `clio-vega/proofs`. **Commit:** `008f8ef5` (full: `008f8ef51529d2194aac19e2e2f5a701488499a9`) — sources, verification code (`proofs/code-1024-q565/`) and this `.tex` are all contained in that single commit, which was made *before* this block was filled in and whose hash was then read off the repository and pasted here. Nothing in this field is deferred to a later action.
Registry: `proofs/registry/rick-q488-converse.json`, nodes under `clio-Q553-vanishing-converse-Q565`.

Abstract

Let $\Psi^C_{(1^r),\theta}(M)$ be the one-head object of my 2026-10-22 alternant expansion, over a set $M = \{w_1, \dots, w_n\}$ of distinct nonzero scalars, with ρ-shifts $m_j = \theta_j + j - 1$ pairwise distinct. On 2026-10-23 I proved the complete criterion for vanishing *in* θ; the companion question (Q565) asked whether the reciprocal-pair locus $w_a w_b = 1$, together with $w_a = \pm 1$, is the whole zero locus *in* M. The answer is no, and the reason is better than a counterexample: *the reciprocal locus is not a vanishing locus at all.* Theorem 8 shows that on $w_a w_b = 1$ the function simply forgets the pair, $\Psi^C_{(1^r),\theta}(M) = \Psi^C_{(1^r),\theta}(M \setminus \{w_a, w_b\})$, and on $w_a^2 = 1$ it forgets the single element. Vanishing is the degenerate end of that descent: it occurs exactly when the descent lands below rank r. This yields the complete structural locus (Theorem 11), a *covering* condition $2\mu + \nu > n - r$ which strictly extends my Proposition `prop:recip` (the case $\mu = 1, \nu = 0, n \le r+1$) to chains of reciprocal pairs; it corrects the threshold at which $w_a = \pm 1$ forces vanishing, which I had stated one step too high; and it predicts exactly which reciprocal polynomials divide the numerator (17/17). The pre-registered conjecture P1 — that there is no cancellation locus for $n > r+1$ — is refuted twice: by a dimension count valid for all $n \ge r+2$ (Theorem 15), and by the certified rational point $M = \{-7, 2, 3, 7, -2\}$, $r = 3$, $\theta = (0, 1, 2)$, which is reciprocal-free with all ten $B(T) \ne 0$ and $\Psi^C = 0$ (Theorem 17). For $r = 1$ the zero locus is determined completely, as the vanishing of a Taylor coefficient of the kernel $H(z) = \prod_t \frac{z w_t - 1}{z - w_t}$ (Theorem 21). Finally I report, as a measured conjecture and not a theorem, a second covering law governed by *negation* pairs $w_b = -w_a$, and a cancellation identity it does not explain.

1 What the existing evidence could not see, and why

Q559's conclusions rest on counts of 424/424, 386/386 and 336/336. The sampler behind them (`code-1023-q559/sampler.py`) draws $w_a =$ `F(randint(2,40),randint(1,7))` and rejects any draw with $|w_a| = 1$, $w_a w_b = 1$ or $w_a = w_b$. My own artifact names the first three exclusions and calls them "hygiene". Two observations about that.

First, as today's brief says: those exclusions are a restatement of the question Q565 asks, so the counts are not weak evidence about the locus in M, they are *no* evidence. That much I had written down in advance.

Second — and this I had *not* written down — the draw is `randint(2,40)/randint(1,7)`, so **every w_a is positive**. Positivity is a fourth exclusion, unnamed in the hygiene note and unnamed in the brief, and it is the one that was actually load-bearing. All three vanishing mechanisms isolated in §7 below need a sign: the alternant mechanism needs $x_i + x_k = 0$, which is impossible for positive arguments (for $\theta = (0, 1, \dots, r-1)$ the alternant ratio is $\prod_{i<k}(x_i+x_k)$, Lemma 19), and the explicit cancellation point of Theorem 17 has two negative entries. Re-running the hygienic sampler restricted to the positive orthant gives 0 zeros in 1800 draws; allowing signs in a pool enriched for $\{v, -v\}$ gives 19 zeros in 540.

So the evidence base was blind on *two* axes, not one, and the second axis is the axis on which the answer lives. I record this because the failure shape is the one I keep meeting: the hygiene rule was written to exclude a degeneracy I understood, and it silently excluded a region I had no hypothesis about. A sampler's *support* is a hypothesis whether or not it is written as one.

2 Setting

Convention 1. $M = \{w_1, \dots, w_n\}$ carries the index order $1 < \cdots < n$. For $r \geq 0$ and $\theta \in \mathbb{Z}^r$ put $m_j = \theta_j + j - 1$. With $\kappa = (1^r)$ (so $\rho \equiv 1$),

$$\Psi^C_{(1^r),\theta}(M) = \sum_{\substack{h=(h_1,\dots,h_r)\\ \text{distinct in } M}} \prod_{j=1}^{r} (1 - X_j^{-2}) X_j^{\theta_j} \prod_{j<k} \frac{1 - \frac{1}{X_j X_k}}{1 - \frac{X_j}{X_k}} \prod_{t\in M,\ t\notin h} \prod_{j=1}^{r} \frac{1 - \frac{1}{w_t X_j}}{1 - \frac{w_t}{X_j}}, \tag{1}$$

$X_j = w_{h_j}$. Write $e = e(M) = \prod_t w_t$. Throughout, M *is admissible* means the w_a are pairwise distinct and nonzero; then every denominator in (1) is nonzero, so Ψ^C is a regular function on the open set

$$U_n = \{w \in \mathbb{C}^n : \ w_a \neq 0,\ w_a \neq w_b \ (a \neq b)\} \subseteq \mathbb{C}^n.$$

The sum (1) is over a permutation-closed index set, so Ψ^C is a *symmetric* function of M; in particular it does not depend on the index order, and for $n < r$ the sum is empty and $\Psi^C = 0$.

I take as given from `2026-10-23-Q559-vanishing-converse.tex`:

Theorem 2 (alternant expansion, 2026-10-22; `clio-Q553-alternant-expansion`)**.** $\Psi^C_{(1^r),\theta}(M) = \sum_{\substack{T\subseteq M\\ |T|=r}} B(T)\, \dfrac{\det(x_i^{m_j})_{i,j=1}^r}{V(T)}$, $V(T) = \prod_{i<k}(x_k - x_i)$, *$x_1, \dots, x_r$ the elements of T in increasing index order, and*

$$B(T) = \prod_{x\in T}(1 - x^{-2}) \prod_{\{x,y\}\subseteq T} \left(1 - \tfrac{1}{xy}\right) \prod_{t\in M\setminus T} \prod_{x\in T} \frac{1 - \frac{1}{w_t x}}{1 - \frac{w_t}{x}}. \tag{2}$$

Theorem 3 (generic non-vanishing, 2026-10-23; `thm:main`)**.** *If $n \geq r$ and the m_j are pairwise distinct then $\Psi^C_{(1^r),\theta} \not\equiv 0$ in $\mathbb{Q}(w_1, \dots, w_n)$.*

Proposition 4 (reciprocal pairs, 2026-10-23; `prop:recip`)**.** *If M is admissible, $w_a w_b = 1$ for some $a \neq b$, and $n \leq r + 1$, then $B(T) = 0$ for every r-subset T, hence $\Psi^C_{(1^r),\theta}(M) = 0$ for every θ.*

Fix throughout, unless said otherwise, θ with the m_j *pairwise distinct*: otherwise Q559's law already gives 0 for every M and the question is empty.

3 The pair form of $B(T)$

Everything below comes from one rewriting. Read (2) as a product over *unordered index pairs*: for $a \le b$ let $\pi_{ab} = w_a w_b - 1$, with $\pi_{aa} = w_a^2 - 1$.

Lemma 5. *For admissible M and $|T| = r$,*

$$B(T) = \frac{N(T)}{D(T)}, \qquad N(T) = \prod_{\substack{a \le b \\ \{a,b\} \cap T \neq \emptyset}} (w_a w_b - 1), \qquad D(T) = e^{\,r} \Big(\prod_{a \in T} w_a \Big) \prod_{a \in T} \prod_{t \notin T} (w_a - w_t),$$

and $D(T) \neq 0$.

Proof. In (2) write $1 - x^{-2} = \frac{x^2-1}{x^2}$, $1 - \frac{1}{xy} = \frac{xy-1}{xy}$ and

$$\frac{1 - \frac{1}{w_t x}}{1 - \frac{w_t}{x}} = \frac{(w_t x - 1)/(w_t x)}{(x - w_t)/x} = \frac{w_t x - 1}{w_t \, (x - w_t)}.$$

The three numerators are $\prod_{a \in T} (w_a^2 - 1)$, $\prod_{\{a,b\} \subseteq T} (w_a w_b - 1)$ and $\prod_{t \notin T} \prod_{a \in T} (w_t w_a - 1)$. A pair $\{a, b\}$ with $a \le b$ contributes to exactly one of these according as $a = b \in T$, or $a \neq b$ with both in T, or $a \neq b$ with exactly one in T; and it contributes to none precisely when $a, b \notin T$. Hence the total numerator is $N(T)$.

For the denominator: the monomial part is $\prod_{a \in T} w_a^2 \cdot \prod_{\{a,b\} \subseteq T} w_a w_b \cdot \prod_{t \notin T} w_t^r = \big(\prod_T w\big)^2 \big(\prod_T w\big)^{r-1} \prod_{t \notin T} w_t^r = \big(\prod_T w\big)^{r+1} \big(e / \prod_T w\big)^r = e^{\,r} \prod_{a \in T} w_a$, and the remaining factors are $\prod_{a \in T} \prod_{t \notin T} (w_a - w_t)$. Admissibility makes both nonzero. □

Verified against the `B()` of Theorem 2 in exact rational arithmetic: 4480/4480 for $n \le 6$, $r \le 4$, 40 random admissible M per cell (`t1_pairform.py`).

Corollary 6. *Let M be admissible and define the* reciprocal support

$$S = S(M) = \{a \in [n] \ : \ w_a w_b = 1 \text{ for some } b \in [n], \ \text{possibly } b = a\}.$$

Then for $|T| = r$: $B(T) \neq 0 \iff T \cap S = \emptyset$.

Proof. By Lemma 5, $B(T) = 0$ iff some pair $\{a, b\}$ with $a \le b$, $w_a w_b = 1$ meets T. If $T \cap S = \emptyset$ then no such pair meets T, since both members of any reciprocal pair lie in S. Conversely if $a \in T \cap S$, take b with $w_a w_b = 1$; the pair $\{a, b\}$ (or $\{a, a\}$) meets T, so $N(T) = 0$. □

Remark 7. S has simple structure. No index lies in two reciprocal pairs: $w_a w_b = w_a w_c = 1$ forces $w_b = w_c$, and $w_a^2 = w_a w_b = 1$ forces $w_b = 1/w_a = w_a$, both excluded by admissibility. So the reciprocal pairs form a partial matching on $[n]$ together with vertex-disjoint loops, and

$$|S| = 2\mu + \nu, \qquad \mu = \#\{a < b : \ w_a w_b = 1\}, \quad \nu = \#\{a : \ w_a = \pm 1\} \ (\le 2).$$

4 The reduction theorem

This is the result that reorganises Q565. Both halves are two lines.

Theorem 8 (reduction)**.** *Let M be admissible and $\theta \in \mathbb{Z}^r$ arbitrary.*

(a) If $w_a^2 = 1$ then $\Psi^C_{(1^r),\theta}(M) = \Psi^C_{(1^r),\theta}\big(M \setminus \{w_a\}\big)$.

(b) If $w_a w_b = 1$ with $a \neq b$ then $\Psi^C_{(1^r),\theta}(M) = \Psi^C_{(1^r),\theta}\big(M \setminus \{w_a, w_b\}\big)$.

Proof. Use Theorem 2. In both cases the alternant factor $\det(x_i^{m_j})/V(T)$ of a term depends only on T, not on M; so it suffices to compare the weights $B(T)$ formed over M and over the smaller set, and to check that the terms which have no counterpart vanish.

(a) If $a \in T$ then the factor $1 - w_a^{-2} = 0$ in (2), so that term is 0. If $a \notin T$ then $t = a$ occurs in the third product of (2), and for each $x \in T$, using $1/w_a = w_a$,

$$\frac{1 - \frac{1}{w_a x}}{1 - \frac{w_a}{x}} = \frac{1 - \frac{w_a}{x}}{1 - \frac{w_a}{x}} = 1.$$

Every other factor of (2) is unchanged when w_a is deleted from M. Hence $B_M(T) = B_{M\setminus\{w_a\}}(T)$ for every $T \subseteq M \setminus \{w_a\}$, and the two sums agree term by term.

(b) If $a, b \in T$ the pair factor $1 - \frac{1}{w_a w_b} = 0$; if exactly one of them lies in T, say $a \in T$, $b \notin T$, the factor of the third product with $t = b$, $x = w_a$ is $1 - \frac{1}{w_b w_a} = 0$. Either way the term vanishes. If $a, b \notin T$ then $t = a$ and $t = b$ both occur in the third product, and for each $x \in T$, using $w_b = 1/w_a$,

$$\frac{1 - \frac{1}{w_a x}}{1 - \frac{w_a}{x}} \cdot \frac{1 - \frac{1}{w_b x}}{1 - \frac{w_b}{x}} = \frac{1 - \frac{1}{w_a x}}{1 - \frac{w_a}{x}} \cdot \frac{1 - \frac{w_a}{x}}{1 - \frac{1}{w_a x}} = 1.$$

Again every other factor is unchanged, so $B_M(T) = B_{M\setminus\{w_a,w_b\}}(T)$ for every T in the smaller set, and the sums agree term by term. □

Remark 9. The two cancellations are the same cancellation. In the pair form of Lemma 5 the third product of (2) contributes, for $t \notin T$ and $x = w_a \in T$, the numerator $\pi_{ta} = w_t w_a - 1$ against the denominator $w_t(w_a - w_t)$; when w_t and some $w_{t'}$ are reciprocal the two numerators $\pi_{ta}\pi_{t'a}$ and the two denominators cancel against each other in pairs. The reciprocal pair acts as a transparent spectator.

Verified in exact rational arithmetic, both sides computed by *both* the head-injection sum (1) and the alternant expansion (Theorem 2): (a) 396/396, (b) 290/290, for $r \leq 3$, $n \leq 6$ and five families of θ including negative entries (`t4_reduction.py`).

Corollary 10 (iterated reduction)**.** *Let M be admissible, S its reciprocal support, and $M^\circ = M \setminus S$. Then M° is admissible and* reciprocal-free *($S(M^\circ) = \emptyset$), $|M^\circ| = n - |S| = n - 2\mu - \nu$, and*

$$\Psi^C_{(1^r),\theta}(M) = \Psi^C_{(1^r),\theta}(M^\circ).$$

Proof. By Remark 7 the reciprocal pairs and loops of M are pairwise disjoint, so S is the disjoint union of μ pairs and ν loops. Apply Theorem 8 once for each, in any order; deleting a reciprocal pair cannot create or destroy another one, again by disjointness. Finally $S(M^\circ) = \emptyset$ because every index belonging to a reciprocal pair of M lies in S and has been deleted. □

5 The structural locus

Theorem 11 (structural locus)**.** *Let M be admissible. Then $B(T) = 0$ for* every *r-subset $T \subseteq M$ if and only if*

$$2\mu + \nu \;>\; n - r,$$

equivalently $|M^\circ| < r$. In that case $\Psi^C_{(1^r),\theta}(M) = 0$ for every θ.

Proof. By Corollary 6, an r-subset with $B(T) \neq 0$ exists iff $[n] \setminus S$ contains r elements, i.e. iff $n - |S| \geq r$. So all $B(T)$ vanish iff $n - |S| < r$, and $|S| = 2\mu + \nu$ by Remark 7. The vanishing of Ψ^C is then immediate from Theorem 2; it also follows from Corollary 10, since $|M^\circ| < r$ makes the sum (1) over M° empty. □

Theorem 11 strictly extends Proposition 4, which is the case $\mu = 1$, $\nu = 0$: there $2\mu + \nu = 2 > n - r$ reads $n \leq r + 1$. Two consequences that Proposition 4 cannot see:

Corollary 12 (chains)**.** *If M contains μ disjoint reciprocal pairs then $\Psi^C_{(1^r),\theta}(M) = 0$ for every θ as soon as $n \leq r + 2\mu - 1$ — however large $n - r$ is. For example $r = 2$, $n = 5 = r + 3$, $M = \{2, \frac{1}{2}, 3, \frac{1}{3}, w\}$ gives $\Psi^C = 0$ for every θ and every admissible w.*

Corollary 13 (the threshold for $w_a = \pm 1$, correcting my own sentence)**.** *If $w_a = \pm 1$ and $n \geq r+1$ then $\Psi^C_{(1^r),\theta}(M)$ need not vanish; a single $w_a = \pm 1$ forces vanishing for all θ exactly when $n \leq r$.*

Proof. $\mu = 0$, $\nu = 1$ in Theorem 11: $1 > n - r$ iff $n \leq r$. (Directly: by Theorem 8(a), $\Psi^C(M) = \Psi^C(M \setminus \{w_a\})$, which is a nonzero rational function as soon as $n - 1 \geq r$, by Theorem 3.) □

Corollary 13 corrects the parenthetical of my Q559 "Open (new)" item, which asked whether the reciprocal locus *together with* $w_a = \pm 1$ is the whole zero locus. As stated, that put $w_a = \pm 1$ on the same footing as $w_a w_b = 1$; in fact it sits one step lower, exactly as today's brief predicted from the 0/0 cancellation in the third product. The brief's pre-registered guess that the two conditions are "one condition on the multiset of indices" is confirmed in the form of Theorem 11: both are instances of $\pi_{ab} = 0$ with $a \leq b$, the diagonal $a = b$ covering one index instead of two. The brief's further guess, that the resulting threshold is a single-pair condition, is *not* confirmed: it is a covering condition, and Corollary 12 shows the covering can be achieved by several pairs at once.

Corollary 14 (which reciprocal polynomials divide the numerator)**.** *Let $n \geq r$ and the m_j be pairwise distinct, and write $\Psi^C_{(1^r),\theta} = p/q$ in lowest terms in $\mathbb{Q}[w_1, \ldots, w_n]$. Then for $a \neq b$,*

$$(w_a w_b - 1) \mid p \iff n \leq r + 1, \qquad \text{and} \qquad (w_a^2 - 1) \mid p \iff n \leq r.$$

Proof. $w_a w_b - 1$ is irreducible, so it divides p iff Ψ^C vanishes identically on the hypersurface $\{w_a w_b = 1\}$. A generic point of that hypersurface is admissible with $\mu = 1, \nu = 0$, and there $\Psi^C(M) = \Psi^C(M \setminus \{w_a, w_b\})$ by Theorem 8(b). The right side is identically zero iff $n - 2 < r$, by Theorem 3 (non-vanishing when $n - 2 \geq r$) and by the empty sum (vanishing when $n - 2 < r$). So the condition is $n \leq r + 1$. The same argument with Theorem 8(a) gives $n - 1 < r$, i.e. $n \leq r$, for $w_a^2 - 1$. □

Corollary 14 is a prediction about symbolic polynomial division, a different instrument from the exact-arithmetic evaluations above. Measured: 17/17 over $1 \leq n \leq 4$, $1 \leq r \leq \min(n, 3)$, $\theta = (0, 1, \ldots, r - 1)$ (`t6_witness.py`). The full factorisations (`t3_factor.py`) agree in detail: at $n = r + 1$ every $(w_a w_b - 1)$ appears (e.g. $n = 3, r = 2$ gives $(w_1 w_2 - 1)(w_1 w_3 - 1)(w_2 w_3 - 1)$ times an irreducible quintic), at $n = r$ the $(w_a^2 - 1)$ appear as well, and at $n \geq r + 2$ no reciprocal factor appears at all.

6 P1 is refuted: there is always a cancellation locus

The brief pre-registered

> **P1.** There is no cancellation locus: for $n > r + 1$ and m_j pairwise distinct, $\Psi^C = 0$ forces every $B(T) = 0$.

with the falsifier "one exhibited M over $\mathbb{Q}$ with $n > r+1$, pairwise distinct m_j, $\Psi^C = 0$, and a nonzero $B(T)$". P1 is false, and in two independent ways.

6.1 A dimension count, valid for every $n \geq r+2$

Theorem 15. *Let $n \geq r+2$ and let the m_j be pairwise distinct. Let $Z = \{w \in U_n : \Psi^C_{(1^r),\theta}(w) = 0\}$ and let $Z_{\mathrm{str}} \subseteq U_n$ be the structural locus of Theorem 11. Then $Z_{\mathrm{str}} \subsetneq Z$ and $\dim Z = n-1 > n-2 \geq \dim Z_{\mathrm{str}}$. In particular there is a point of U_n at which $\Psi^C = 0$ and some $B(T) \neq 0$.*

Proof. Write $\Psi^C = p/q$ in lowest terms in $\mathbb{Q}[w]$ (a UFD). Ψ^C is regular on U_n (Convention 1); since p, q are coprime, q cannot vanish anywhere on U_n, so $Z = V(p) \cap U_n$. By Theorem 3, $p \neq 0$.

$Z \neq \emptyset$: put $d = n - r + 1$; then $3 \leq d \leq n$ (since $r \geq 1$). If d is even take $\mu = d/2$ disjoint reciprocal pairs and $\nu = 0$; if d is odd take $\mu = (d-1)/2$ pairs and one loop $w_c = 1$ ($\nu = 1$). Either way $2\mu + \nu = d \leq n$, so the pairs and the loop fit on disjoint indices; realise the pairs by $\{v_i, 1/v_i\}$ with the v_i generic and the remaining coordinates generic, so the point is admissible. Then $2\mu + \nu = d > n - r$, so $\Psi^C = 0$ there by Theorem 11. (Corrected 2026-11-12 after Rick's referee read, UID 903: the earlier text took $\lceil d/2 \rceil$ pairs "since $2\mu \leq n$", which fails for $r = 1$ and n odd, where $2\lceil d/2 \rceil = n + 1$.) Hence p is nonconstant, $V(p)$ is a hypersurface, and Z is a nonempty open subset of it, so $\dim Z = n - 1$.

$\dim Z_{\mathrm{str}} \leq n - 2$: by Theorem 11 and Remark 7, Z_{str} is the union, over partial matchings with loops $(\mathcal{P}, \mathcal{L})$ on $[n]$ satisfying $2|\mathcal{P}| + |\mathcal{L}| > n - r$, of the sets $Y_{\mathcal{P},\mathcal{L}} = \{w_a w_b = 1 \ \forall \{a,b\} \in \mathcal{P};\ w_c^2 = 1 \ \forall c \in \mathcal{L}\} \cap U_n$. Each such set is cut out by $|\mathcal{P}| + |\mathcal{L}|$ equations in *disjoint* groups of variables, so it is a graph over the free coordinates and $\dim Y_{\mathcal{P},\mathcal{L}} = n - |\mathcal{P}| - |\mathcal{L}|$ exactly. Minimising $|\mathcal{P}| + |\mathcal{L}|$ subject to $2|\mathcal{P}| + |\mathcal{L}| \geq n - r + 1$ gives $|\mathcal{P}| + |\mathcal{L}| \geq \lceil (n - r + 1)/2 \rceil \geq 2$ because $n - r + 1 \geq 3$. Hence every piece has dimension $\leq n - 2$, and so does the finite union.

Therefore $Z \not\subseteq Z_{\mathrm{str}}$, while $Z_{\mathrm{str}} \subseteq Z$ by Theorem 11. Any point of $Z \setminus Z_{\mathrm{str}}$ has $\Psi^C = 0$ and, by Theorem 11, some $B(T) \neq 0$. □

Remark 16 (where the brief's proposed test would have misled me). The brief proposed to decide P1 by factoring the numerator, with the rule: "if every irreducible factor is of the form $(w_a w_b - 1)$ or $(w_a^2 - 1)$, P1 is confirmed *and I have the proof*, because a product of those factors is exactly the structural locus." That inference is wrong, and Theorem 15 is why. A product of reciprocal factors cuts out $\bigcup_{a \leq b} \{w_a w_b = 1\}$, which has codimension 1; the structural locus has codimension $\lceil (n - r + 1)/2 \rceil$, which is ≥ 2 as soon as $n \geq r + 2$. "There is a reciprocal pair" and "every $B(T)$ vanishes" are two different conditions that agree only for $n \leq r + 1$. So even the outcome the brief called a confirmation would have been a refutation. This is the same fault the brief warned me about one paragraph earlier — one word over two mechanisms — committed in the sentence that set up the test.

6.2 A certified rational witness

Theorem 17. *Let $M = \{-7, 2, 3, 7, -2\}$ (so $n = 5$), $r = 3$, $\theta = (0, 1, 2)$, hence $m = (0, 2, 4)$. Then M is admissible with $n = r + 2$, the m_j are pairwise distinct, M is reciprocal-free ($S = \emptyset$, so $n - |S| = 5 \geq r$), all ten $B(T) \neq 0$, and $\Psi^C_{(1^3),(0,1,2)}(M) = 0$.*

Computed in exact rational arithmetic by both implementations (`t9_certify.py`). Four of the

ten terms are nonzero, and they cancel in two pairs:

$$\begin{array}{lllll} T=\{-7,2,3\}: & B=\frac{232375}{4941258}, & \text{alt}=100, & \text{term}=+\frac{11618750}{2470629} \\ T=\{3,7,-2\}: & B=-\frac{232375}{2470629}, & \text{alt}=50, & \text{term}=-\frac{11618750}{2470629} \\ T=\{-7,3,-2\}: & B=-\frac{11171875}{133413966}, & \text{alt}=36, & \text{term}=-\frac{22343750}{7411887} \\ T=\{2,3,7\}: & B=\frac{446875}{66706983}, & \text{alt}=450, & \text{term}=+\frac{22343750}{7411887} \end{array}$$

The pairing is by the involution of M that exchanges $7 \leftrightarrow -7$ and $2 \leftrightarrow -2$ and fixes 3. This is the pre-registered falsifier, met exactly.

Remark 18 (the witness came out of a control that "failed"). Control 1 of the brief was: the Q559 hygienic sampler must report no zeros, and "a zero here means my engine regressed". My re-implementation of that hygiene reported 19 zeros in 540 draws. The engine had not regressed: as §1 explains, I had reproduced the three *named* exclusions and not the unnamed one (positivity), and the 19 zeros all contain a negative w_a. Classifying them gave 18 of mechanism M2 and exactly one of mechanism M3 — which is Theorem 17. The third time running, the control I was told to expect green from was the instrument that found the result. I now think the lesson is sharper than "diagnose a failing control": a control states a prediction, and when it fires, the *first* thing to check is whether I reproduced the instrument or only its description.

7 Three mechanisms, not two

Theorem 2 writes Ψ^C as $\sum_T B(T) \cdot A(T)$ with $A(T) = \det(x_i^{m_j})/V(T)$. A zero can therefore arise in three ways, and the brief's framing of Q565 as "structural versus cancellation" is one split short.

M1, structural. Every $B(T) = 0$. By Theorem 11 this is exactly $2\mu + \nu > n - r$. By Theorem 8 it is not really a vanishing mechanism at all but the degenerate end of a restriction.

M2, alternant. Every term with $B(T) \neq 0$ has $A(T) = 0$. Since $A(T) = s_\lambda(T)$ for the partition $\lambda_j = m_{r+1-j} - (r-j)$, this is a Schur-vanishing condition on T; it is invisible to any argument about B. Smallest witness: $n = r = 2$, $\theta = (0,1)$ so $m = (0,2)$, $A(T) = x_1 + x_2$, $M = \{3,-3\}$: $B(M) = \frac{640}{729} \neq 0$, no reciprocal pair, $\Psi^C = 0$ (`t6_witness.py`).

M3, cancellation. Nonzero terms sum to zero. Witness: Theorem 17.

M2 is not a curiosity. For the natural choice $\theta = (0, 1, \ldots, r-1)$, i.e. $m_j = 2(j-1)$, the alternant is a Vandermonde in the squares and

Lemma 19. *If $m_j = 2(j-1)$ for $1 \le j \le r$ then* $A(T) = \dfrac{\det(x_i^{2(j-1)})}{\prod_{i<k}(x_k - x_i)} = \displaystyle\prod_{1 \le i < k \le r} (x_i + x_k).$

Proof. $\det(x_i^{2(j-1)})_{i,j}$ is the Vandermonde determinant in $x_1^2, \ldots, x_r^2$, namely $\prod_{i<k}(x_k^2 - x_i^2)$. Divide by $\prod_{i<k}(x_k - x_i)$. □

Verified 240/240 for $r \le 4$ (`t9_certify.py`). So for this θ the term of T dies iff T contains both members of a *negation* pair $w_b = -w_a$, and the same covering argument as Theorem 11 applies to $(w_a + w_b)$ in place of $(w_a w_b - 1)$:

Proposition 20 (negation covering law)**.** *Let M be admissible, $\theta = (0, 1, \ldots, r-1)$, and let λ be the number of pairs $a < b$ with $w_b = -w_a$ (these pairs are disjoint, since $w_b = -w_a = -w_c$ forces $w_b = w_c$). Then $A(T) = 0$ for every r-subset T if and only if $r > n - \lambda$.*

Proof. By Lemma 19, $A(T) \neq 0$ iff T contains at most one member of each negation pair. A largest such T takes one member from each of the λ pairs together with all $n - 2\lambda$ unpaired elements, so has size $n - \lambda$. Hence such a T of size r exists iff $r \leq n - \lambda$. □

Measured 72/72 over $\lambda \leq 3$, $f \leq 3$ unpaired elements, all r (`t9_certify.py`); and it accounts for exactly the 18 M2 zeros of the failed control.

Note the duality: B contributes $\prod(w_a w_b - 1)$ over pairs *meeting* T, and at this θ the alternant contributes $\prod(w_a + w_b)$ over pairs *inside* T. The two covering laws have identical shape and opposite incidence.

8 $r = 1$: the zero locus, completely

For $r = 1$ the question can be answered outright, and the answer is a Taylor coefficient of the kernel that already governs this object.

Theorem 21. *Let M be admissible, $r = 1$, $\theta = (m)$ with $m \in \mathbb{Z}$, and*

$$H(z) = \prod_{t=1}^{n} \frac{zw_t - 1}{z - w_t} = \sum_{k \geq 0} h_k \, z^{-k} \quad (\text{at } z = \infty) = \sum_{k \geq 0} \tilde{h}_k \, z^{k} \quad (\text{at } z = 0).$$

Then

$$\Psi^{C}_{(1),(m)}(M) = \frac{1}{e} \cdot \begin{cases} h_m, & m \geq 1, \\ e - e^{-1}, & m = 0, \\ -\tilde{h}_{-m}, & m \leq -1, \end{cases} \qquad h_k = \sum_{j=0}^{\min(k,n)} (-1)^j e_{n-j}(w) \, h_{k-j}(w),$$

e_i and h_i the elementary and complete homogeneous symmetric polynomials. Consequently the zero locus of $\Psi^{C}_{(1),(m)}$ in U_n is $\{h_m = 0\}$ for $m \geq 1$, $\{e^2 = 1\}$ for $m = 0$, and $\{\tilde{h}_{-m} = 0\}$ for $m \leq -1$.

Proof. For $r = 1$, (1) reads $\Psi^{C}_{(1),(m)} = \sum_a (1 - w_a^{-2}) w_a^m \prod_{s \neq a} \frac{1 - \frac{1}{w_s w_a}}{1 - \frac{w_s}{w_a}}$. Using $\frac{1 - \frac{1}{w_s w_a}}{1 - \frac{w_s}{w_a}} = \frac{w_s w_a - 1}{w_s (w_a - w_s)}$ and $\prod_{s \neq a} w_s = e / w_a$,

$$\Psi^{C}_{(1),(m)} = \frac{1}{e} \sum_a (w_a^2 - 1) \, w_a^{m-1} \prod_{s \neq a} \frac{w_a w_s - 1}{w_a - w_s} = \frac{1}{e} \sum_a \operatorname*{Res}_{z = w_a} \left[z^{m-1} H(z) \right],$$

because $H(z) = \frac{zw_a - 1}{z - w_a} \prod_{s \neq a} \frac{zw_s - 1}{z - w_s}$ has residue $(w_a^2 - 1) \prod_{s \neq a} \frac{w_a w_s - 1}{w_a - w_s}$ at the simple pole $z = w_a$, and z^{m-1} is regular and nonzero there. The function $z^{m-1} H(z)$ is rational with poles only at $w_1, \ldots, w_n$, at 0 (if $m \leq 0$) and at ∞; the sum of all its residues on $\mathbb{P}^1$ is zero, so

$$\sum_a \operatorname*{Res}_{z = w_a} = - \operatorname*{Res}_{z = 0} - \operatorname*{Res}_{z = \infty} = -\tilde{h}_{-m} \, [m \leq 0] + h_m \, [m \geq 0],$$

since $\operatorname{Res}_{z=0} z^{m-1} H = [z^{-1}] \big(z^{m-1} \sum \tilde{h}_k z^k \big) = \tilde{h}_{-m}$ when $m \leq 0$ and 0 otherwise, and $\operatorname{Res}_{z=\infty} f = -[z^{-1}] f$ gives $-h_m$ when $m \geq 0$ and 0 otherwise. At $m = 0$ both contribute, and $h_0 = e$, $\tilde{h}_0 = e^{-1}$. Finally $H(z) = e \prod_t \frac{1 - (zw_t)^{-1}}{1 - w_t / z}$, so $h_k = e \sum_j (-1)^j e_j(w^{-1}) h_{k-j}(w)$ and $e_j(w^{-1}) = e_{n-j}(w)/e$. □

Verified 540/540 for $n \leq 5$, $-3 \leq m \leq 5$ against (1) (`t5_closedform.py`).

Corollary 22. $\Psi^C_{(1),(0)}(M) = 1 - e(M)^{-2}$ *for every* n *and every admissible* M*: it depends on* M *only through the product* e*. Its zero locus is* $\{e = \pm 1\}$*, which for* $n \geq 3$ *contains reciprocal-free rational points — e.g.* $M = \{2, 3, \frac{1}{6}\}$ *with* $n = 3 = r + 2$*, where all three* $B(T) \neq 0$ *and* $\Psi^C = 0$*.*

Verified 14/14 for $n \leq 7$ in two families, and the witness $M = \{2, 3, \frac{1}{6}\}$ certified by both implementations (`t7_witnesses.py`). Corollary 22 is a second, independent refutation of P1, at $r = 1$.

Remark 23. Theorem 21 also settles the shape of the locus. At $r = 1$, $n = 3$, $m = 1$ the numerator is $e_3h_1 - e_2 = w_1w_2w_3(w_1+w_2+w_3) - (w_1w_2+w_1w_3+w_2w_3)$, irreducible of degree 4 (`t3_factor.py`); restricted to $w_1w_2 = 1$ it becomes $w_3^2 - 1$, which is $e \cdot \Psi^C_{(1),(1)}(\{w_3\})$ — Theorem 8(b) visible in the factorisation. So for $n \geq r + 2$ the zero locus is an irreducible hypersurface having nothing to do with reciprocal pairs, exactly as Theorem 15 requires.

9 What Q565 now asks

Corollary 10 replaces Q565 by a strictly smaller question.

Corollary 24. *For admissible* M *with reciprocal support* S *and* $M^\circ = M \setminus S$*,*

$$\Psi^C_{(1^r),\theta}(M) = 0 \iff |M^\circ| < r \quad \text{or} \quad \Psi^C_{(1^r),\theta}(M^\circ) = 0,$$

and the second alternative is a condition on a reciprocal-free *set.*

So the answer to the question as my Q559 artifact posed it is: *no* — but not because the reciprocal locus is too small a guess. It is because the reciprocal locus is the wrong *kind* of object. The zero locus proper lives entirely on the reciprocal-free stratum, and what the reciprocal locus does is fibre the problem over smaller n. Determining the zero locus on the reciprocal-free stratum is complete for $r = 1$ (Theorem 21) and open for $r \geq 2$; Theorems 15 and 17 show it is nonempty and genuinely a cancellation phenomenon.

10 A measured conjecture I did not prove

Sweeping the negation configurations gave a threshold sharper than Proposition 20, and one that subsumes it.

Fix $\theta = (0, 1, \ldots, r-1)$ and let M be admissible and reciprocal-free, consisting of λ disjoint negation pairs $\{v_i, -v_i\}$ together with f unpaired elements, so $n = 2\lambda + f$. By Lemma 19 a term survives only if T takes at most one member of each negation pair; since T can draw at most f unpaired elements, a surviving T of size r must *touch at least* $r - f$ *distinct negation pairs.* So $r \geq f + 2$ says exactly: every surviving T touches two or more pairs.

Conjecture 25 (**refuted**; see the note after it)**.** In the situation above,

$$\Psi^C_{(1^r),(0,1,\ldots,r-1)}(M) = 0 \qquad \Longleftrightarrow \qquad r \;\geq\; f + 2.$$

Refuted (added 2026-11-12). The "⇒" direction is false, and the witness is already in this note: $M = \{2, 3, \frac{1}{6}\}$ of Corollary 22 has $\lambda = 0$, $f = 3$, $r = 1 < f + 2$, is admissible and reciprocal-free, and has $\Psi^C_{(1),(0)}(M) = 1 - e^{-2} = 0$. Four further reciprocal-free witnesses, one inside the swept cell $\lambda = 3$, $f = 0$, $r = 1$, and the corrected statement (identical vanishing iff $r \geq f + 2 - \delta$) are Theorem 18 and Corollary 21 of `2026-10-24-c2-Q576-transversal-cancellation.tex`. The

sweep below could not see this: $\{e = \pm 1\}$ has codimension 1 in the stratum. (Pointed out by Rick, referee read UID 903.) The original text follows unchanged.

Measured 26/26 cells, both directions, for $\lambda \le 4$, $f \le 3$, $n \le 7$ and $r \in \{f+1, f+2, f+3\}$ (the threshold and its two neighbours), 10 draws per cell, both implementations agreeing on every draw (`t12_sharp.py`); and 14 further cells over the full range of r (`t11_grid.py`). The boundary is clean: $r = f + 1$ gave 0 zeros in every cell, so the threshold is not an artefact of the sweep.

The conjecture unifies the two mechanisms. Writing $\lambda + f = n - \lambda$:

- for $r > n - \lambda$, i.e. $r > \lambda + f$, no T survives and the vanishing is Proposition 20 — *proved*;
- for $f + 2 \le r \le \lambda + f$ there are $\sum_k \binom{\lambda}{k} 2^k \binom{f}{r-k}$ surviving terms (choose k pairs to touch, one member from each, and $r - k$ unpaired elements; k runs over $\max(0, r - f) \le k \le \min(\lambda, r)$), all nonzero, and they cancel — *conjectural*. This window is nonempty iff $\lambda \ge 2$, which is why $\lambda = 1$ shows no cancellation at all.

Theorem 17 is the cell $\lambda = 2$, $f = 1$, $r = 3$, where the window is a single value of r and the four surviving terms cancel in two pairs matched by the involution reversing one negation pair. That pairing is surely the proof, but I have not found the sign bookkeeping that makes it work for general λ, and I record the statement with its measured domain rather than dressing a guess as an argument.

11 Verification

All arithmetic is exact (`fractions.Fraction`) or symbolic (SymPy); no floating point anywhere, since a float Ψ^C near a reciprocal locus reads as zero. Two independent implementations of Ψ^C are used throughout: the head-injection sum (1) and the alternant expansion (Theorem 2). Code in `proofs/code-1024-q565/`.

claim	test	domain	result
Lemma 5 (pair form of B)	`t1`	$n \le 6$, $r \le 4$, 40 M/cell	4480/4480
Theorem 11 ($2\mu + \nu > n - r$)	`t2`	$n \le 6$, all $r \le n$, 8 configs	132/132 cells
two implementations agree	`t2`	same	132/132 cells
Theorem 8(a), $w_a = \pm 1$	`t4`	$r \le 3$, $n \le 6$, 5 θ-families	396/396
Theorem 8(b), $w_a w_b = 1$	`t4`	same	290/290
Corollary 10 (iterated, chains)	`t4`	$r \le 3$, $\mu \le 3$, $\nu \le 1$	30/30
Corollary 14 (divisibility)	`t6`	$n \le 4$, $r \le \min(n, 3)$	17/17
Theorem 21 ($r = 1$ closed form)	`t5`	$n \le 5$, $-3 \le m \le 5$	540/540
Corollary 22 ($1 - e^{-2}$)	`t7`	$n \le 7$, 2 families	14/14
Lemma 19 (alternant $= \prod(x_i + x_k)$)	`t9`	$r \le 4$, 60 draws/cell	240/240
Proposition 20 (negation covering)	`t9`	$\lambda \le 3$, $f \le 3$, all r	72/72
Conjecture 25 ($r \ge f+2$)	`t12`	$\lambda \le 4$, $f \le 3$, $r = f+1, f+2, f+3$	26/26 cells
— full r-range grid	`t11`	$\lambda \le 3$, $f \le 3$, all r	14/14 cells
— transversal cells	`t10`	$\lambda \in \{2, 3\}$, $f \le 2$	25/25 each
Controls.			
negative: positive hygienic sampler	`t8`	$r \le 3$, $r \le n \le r + 2$	0 zeros / 1800
positive: `prop:recip` regime fires	`t2`	$n \le r + 1$, planted pair	all 0
separating: $n = r + 2$, avoidable pair	`t2`	$r = 2, n = 4$; $r = 1, n = 3$	$\Psi^C \ne 0$
negative control as re-implemented	`t8`	sign-enriched pool	19/540, all diagnosed

The separating control of the brief is the decisive cell: plant $w_a w_b = 1$ with $n = r + 2$, both members avoidable. Structural vanishing predicts $\Psi^C \ne 0$; a cancellation locus containing the whole

reciprocal hypersurface would predict it could be 0. Measured $\neq 0$, and Theorem 8(b) explains the value: it is Ψ^C over the two-element-smaller set.

12 Grades

- `proved`. Lemma 5, Theorem 8, Corollary 10, Theorem 11, Corollaries 12, 13, 14, Theorem 15, Lemma 19, Proposition 20, Theorem 21, Corollaries 22, 24. Every step is an identity in $\mathbb{Q}(w_1, \ldots, w_n)$ or a dimension count, with no external input. Standing on Theorem 2 (`proved` 2026-10-22) and Theorem 3 (`proved` 2026-10-23); both are mine, and Theorem 2 is re-verified here at 4480/4480 through Lemma 5.

- `computed`. Theorem 17 — a single exhibited point, verified exactly by two implementations. It is a counterexample, so one point is logically sufficient; I grade it `computed` rather than `proved` because its content is an arithmetic evaluation, not an argument. Nothing else rests on it: Theorem 15 refutes P1 independently.

- `conjectured`. Conjecture 25, with the measured domain stated. Also open: the zero locus on the reciprocal-free stratum for $r \geq 2$, and whether $\Psi^C_{(1^r),\theta}$ admits a determinantal formula in the $r = 1$ data of Theorem 21 — the four natural 2×2 ansätze fail ($\leq 1/72$, and the hits are $\Psi^C = 0$ cells, i.e. $0 = 0$; `t5`, `t6`).

- **Retracted.** The parenthetical "(together with $w_a = \pm 1$)" in the Q559 "Open (new)" item is withdrawn: by Corollary 13 that condition forces vanishing only for $n \leq r$, not alongside $w_a w_b = 1$ at $n \leq r + 1$.

- **No external citations.** There are none in this note, hence no locator obligations and no extraction-level obligations.

13 Open questions arising

- **Q565 residue.** Determine the zero locus of $\Psi^C_{(1^r),\theta}$ on the reciprocal-free stratum for $r \geq 2$. By Corollary 24 this is all that remains of Q565.

- **Conjecture 25**: prove the cancellation in the window $f + 2 \leq r \leq \lambda + f$. The involution reversing one negation pair pairs the surviving terms; the missing step is the sign.

- **A contour formula.** Combining Lemma 5 with the residue calculus of Theorem 21 gives, formally,

$$\Psi^C_{(1^r),\theta}(M) = \frac{(-1)^{\binom{r}{2}}}{r!\, e^r} \frac{1}{(2\pi i)^r} \oint \cdots \oint \det\left(z_i^{m_j}\right) \prod_{i<k} \frac{z_k - z_i}{z_i z_k - 1} \prod_j \frac{H(z_j)}{z_j}\, dz,$$

contours around $w_1, \ldots, w_n$ only. I derived the kernel by matching residues against Theorem 2 but have *not* verified it, symbolically or numerically, and it is recorded here only as the next thing to check. The factor $\prod_{i<k} \frac{z_k - z_i}{z_i z_k - 1}$ is the type-C Gaudin kernel, which is presumably why the free-fermionic determinant ansätze fail.